



Coupled modelling of turbulent forced thermal convection in anisotropic porous media

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The momentum dispersion model for flows in isotropic porous media has been validated and successfully applied by Rao & Jin (2022, *J. Fluid Mech.*, vol. 937, A17). However, the anisotropic coupled models concerning heat–fluid–solid interactions in turbulent forced convection requires further development. This research proposes various anisotropic physical coefficient tensors to model the total drag R_i , interphase energy resistance H , momentum dispersion and thermal dispersion accounting for both anisotropic and isotropic scenarios. The effective physical coefficients of the Darcy–Forchheimer equation regarding R_i are adapted to accommodate anisotropy. The heat transfer coefficient h between the solid and fluid, despite being a scalar, is also required to depend on the local flow direction in anisotropic cases. Two scaling laws of h with respect to a local Reynolds number Re_K are found: $h \sim Re_K^2$ for the Darcy regime, and $h \sim Re_K^{1/2}$ for the Forchheimer regime, with a transition at $Re_K \sim 1$. The influence of momentum and thermal dispersions, along with the modelling errors of R_i and H originating from heterogeneity, are approximated using a second-order pseudo-stress tensor and a pseudo-flux vector, respectively. The effective viscosity and thermal diffusivity tensors are simplified into longitudinal and transverse components using tensor symmetries, and are assumed to rely mainly on another local Reynolds number Re_d . Both components of the effective viscosity are positive in isotropic cases, whereas the longitudinal component may be negative in anisotropic cases, mainly serving as a compensation of overestimated drag. The coupled models are applied to simulate turbulent forced thermal convection in porous media with one or two length scales across a wide range of Reynolds numbers. The comparisons with direct numerical simulations results imply that the coupled macroscopic models can accurately predict not only statistically stationary distributions but also real-time changes in velocity and temperature.

Key words: convection in porous media, turbulence modelling, turbulent convection

1. Introduction

The study of turbulent forced thermal convection in porous media is essential due to its wide-ranging applications in engineering and environmental systems (Nield & Bejan 2006; Wood, He & Apte 2020), including the design of heat exchangers, geothermal energy extraction, and cooling systems for electronic devices (Rebhi *et al.* 2022). Turbulent forced convection enhances heat transfer by actively driving fluid through the porous structure, thereby optimising thermal performance in systems where heat transfer alone is insufficient. Furthermore, it plays a crucial role in subsurface processes such as groundwater remediation and carbon sequestration, where a comprehensive understanding of heat and mass transfer in porous matrices can significantly influence environmental sustainability. As industries increasingly demand higher energy efficiency, investigating turbulent forced thermal convection within porous media becomes imperative for effective heat management.

As an efficient substitute for the demanding direct numerical simulations (DNS) for simulating turbulent forced thermal convection in porous media, macroscopic coupled equations can be derived through time and volume averaging of the momentum and energy equations (Pope 2001; de Lemos 2005; Rao & Zhang 2025*b*). The averaging volumes are defined as representative elementary volumes (REV)s, which are appropriately sized to accurately capture the pore-scale properties of the entire volume under consideration. When volume averaging is performed without time averaging of the momentum and energy equations, the total drag, spatial momentum dispersion term, interphase energy resistance and thermal dispersion remain unknown (Nield & Bejan 2006; Rao, Kuznetsov & Jin 2020). To close the macroscopic coupled equations, modelling of these unknown terms is necessary.

Typically, the total drag is modelled using the Darcy–Forchheimer equation (Kuznetsov & Kuznetsov 2017; Lasseux, Valdés-Parada & Bellet 2019; Aghaei-Jouybari *et al.* 2024). The Darcy equation is applicable when the flow velocity is sufficiently low, while the Forchheimer equation modifies the Darcy equation to account for turbulent flow conditions (Irmay 1958; Beavers, Sparrow & Rodenz 1973; Masuoka & Takatsu 1996). An extension to the Darcy–Forchheimer equation is the Brinkman equation (Tam 1969; Givler & Altobelli 1994), which accounts for the effects of spatial momentum dispersion by introducing a Laplacian term of the macroscopic velocity gradient in the macroscopic momentum equation (Nield & Bejan 2006; Valdes-Parada, Ochoa-Tapia & Alvarez-Ramirez 2007; Rao *et al.* 2020; Rao 2022).

In the context of heat transfer between solid and fluid phases, the local thermal equilibrium (LTE) assumption of both phases having identical temperature is a widely utilised model for analysing such systems (Kaviany 2012). This assumption posits that the temperatures of both the solid and fluid phases can be represented by a single value, thereby facilitating theoretical and numerical investigations. However, the LTE assumption oversimplifies the complexities inherent in heat transfer. Consequently, there has been an increasing focus on the local thermal non-equilibrium model (Nakayama *et al.* 2001; Quintard, Cherblanc & Whitaker 2001; Saito & de Lemos 2005; Davit *et al.* 2010), which has garnered considerable attention in both theoretical and numerical research by accounting for the distinct temperatures of the fluid and solid phases in porous media (Saito & De Lemos 2006). An important extension of this concept is the effective thermal conductivity of the porous medium (Samantray, Karthikeyan & Reddy 2006), which accounts for thermal dispersion, molecular diffusion and turbulent convection (Braga & de Lemos 2005).

However, there is a notable deficiency in the availability of coupled models for turbulence and heat transfer that account for both anisotropic and isotropic porous media.

In this paper, we propose models that employ various necessary anisotropic tensors to incorporate total drag, effective viscosity, interphase energy resistance and effective thermal diffusivity. The models are validated against DNS results across a wide range of Reynolds numbers in porous media characterised by one or two length scales. Ultimately, macroscopic coupled equations are effectively utilised to predict statistically stationary distributions and real-time changes in both velocity and temperature.

2. Establishment of anisotropic and time-dependent macroscopic equations

2.1. Theoretical framework

Based on the theorem of local volumetric average, the macroscopic equations governing flows in porous media were often derived by applying both volume and time averaging to the Navier–Stokes equations (de Lemos 2012*a,b*). In order to enable the capability of real-time prediction, our approach employs only volume averaging, as demonstrated in (2.1) below. We assume that the fluid is incompressible, and include a heat source term, leaving the density constant despite variations in temperature. The solid phase is static and has a temperature field T_s whose evolution usually depends on extra solid properties. However, it can be found that the complexities of heat transfer within porous media primarily reside in the fluid phase, either explicitly or implicitly. Therefore, this study focuses on thermal convection within the fluid phase, and only displays the solid-phase energy equation formally. Nevertheless, the modelling framework in this study is applicable to a wide range of geometries and coarse-grained thermodynamic properties of the solid phase. This most crucially requires the incorporation of heat exchange between solid and fluid phases, which involves the local solid–fluid temperature difference. The equations are

$$\partial_t(\phi\langle u_i \rangle^f) = 0, \quad (2.1a)$$

$$\partial_t(\phi\langle u_i \rangle^f) + \partial_j(\phi\langle u_i \rangle^f \langle u_j \rangle^f) = -\partial_i(\phi\langle p \rangle^f)/\rho + \phi g_i - \phi \hat{R}_i + \partial_j \tilde{\tau}_{ij}, \quad (2.1b)$$

$$\partial_t(\phi\langle T_f \rangle^f) + \partial_i(\phi\langle u_i \rangle^f \langle T_f \rangle^f) = \phi \dot{q} - \phi \hat{H} + \partial_i \tilde{J}_i, \quad (2.1c)$$

$$\partial_t(\phi\eta_c\langle T_s \rangle^s) = \phi\eta_c\dot{q}^s + \phi \hat{H} + \partial_i \tilde{J}_i^s, \quad (2.1d)$$

where porosity ϕ represents the fraction of void space to total volume in porous media. Also, $\langle \varphi \rangle^f$ and $\langle \varphi \rangle^s$ denote the intrinsic average values of any fluid and solid field within each REV. The spatial deviation $^f\varphi = \varphi - \langle \varphi \rangle^f$ is defined as the difference between the microscopic value and the intrinsic volume-averaged value. In (2.1*b*), g_i is an applied constant body force, $\hat{R}_i$ is the Darcy–Forchheimer drag force, $\tilde{\tau}_{ij}$ is a second-order pseudo-stress tensor, T_f is the temperature for the fluid phase, $\dot{q}$ is a heat source term, $\hat{H}$ is the interphase energy resistance, and $\tilde{J}_i$ is pseudo-flux vector. Terms in the formal solid energy equation (2.1*d*), including the REV solid–fluid heat capacity ratio $\eta_c = (1 - \phi)\rho^s c_p^s / (\phi\rho c_p)$ (volume ratio times the volumetric isobaric heat capacity ratio), the solid heat source term $\dot{q}^s$, and the solid-phase pseudo-flux vector $\tilde{J}_i^s$, can be arbitrary and are thus not specified in the present work. The dimensions of the variables include length, time, mass and temperature, which are not explicitly defined in this context.

Far away from wall boundaries, the total drag R_i is decomposed into two components: the surface integrals of the intrinsic average pressure and viscous forces R_i^γ , and the surface integrals of the spatial deviations of pressure and viscous forces $\tilde{R}_i$. Here, R_i^γ can be approximated to zeroth order of corresponding values multiplied by the negative porosity gradient. Given that the porosity is considered a constant value for macroscopic simulation, R_i^γ equals zero. Consequently, the total drag $R_i = \tilde{R}_i$, and is approximated

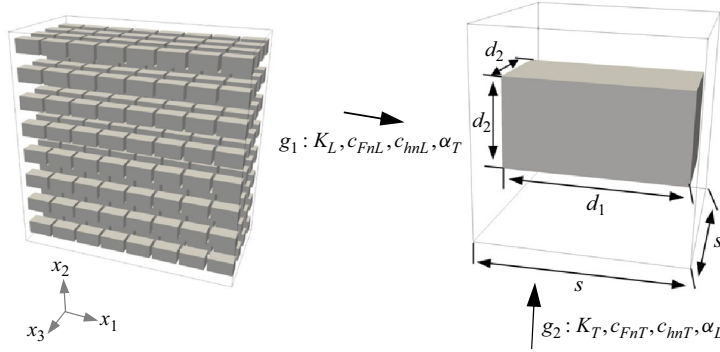


Figure 1. A schematic of a transversely isotropic porous matrix made up of a cuboid, with ratio 2.0, and $\phi = 0.81$.

by the Darcy–Forchheimer equation $\hat{R}_i$, as shown below (Whitaker 1986; Breugem, Boersma & Uittenbogaard 2006):

$$R_i = (\phi \Delta V)^{-1} \int_A (\hat{n}_i p / \rho - v \hat{n}_j \partial_j u_i) dS \approx \hat{R}_i = v K^{-1} \left(\sum_{n=0}^{\infty} c_{Fn} Re_K^n \right) u_{Di}, \quad (2.2)$$

which adopts scalar permeability K and Taylor expansion coefficients c_{Fn} to account for the total drag exclusively caused by isotropic porous media, with c_{F0} being 1. Here, ΔV denotes the volume of one REV, A denotes the fluid–solid interfacial area, and $\hat{n}_i$ (in surface integrals) denotes the normal direction from the fluid to porous matrix at the solid–fluid interface. To extend this model for anisotropic porous media, we should generalise K and c_{Fn} into tensors that reflect anisotropic dependence of $\hat{R}_i$ on flow direction and local Reynolds number $Re_K = |u_D| \sqrt{|K_{ij}|} / \nu$, where $u_{Di} = \phi \langle u_i \rangle^f$:

$$\hat{R}_i = v K_{ik}^{-1} \left(\sum_{n=0}^{\infty} c_{Fnkj} Re_K^n \right) u_{Dj}. \quad (2.3)$$

Here, c_{Fnkj} represent the coefficients of the Taylor series, with c_{F0kj} being a unit matrix. For simplicity, in the present work we consider the porous matrices that are isotropic in the planes perpendicular to a specific direction $\tilde{n}$ (transversely isotropic, as sketched in figure 1), which represents a wide range of anisotropic porous media. With additional symmetry required by energy conservation, K_{ij} and c_{Fnkj} can be decomposed as follows:

$$K_{ij} = K_L \tilde{n}_i \tilde{n}_j + K_T (\delta_{ij} - \tilde{n}_i \tilde{n}_j), \quad c_{Fnij} = c_{FnL} \tilde{n}_i \tilde{n}_j + c_{FnT} (\delta_{ij} - \tilde{n}_i \tilde{n}_j). \quad (2.4)$$

The characteristic direction $\tilde{n}_i$ is associated with the detailed geometry of the porous matrix, and δ_{ij} denotes unit matrix. The permeability K_{ij} indicates the capacity of the porous medium to facilitate fluid flow, and $|K_{ij}| = [3/(K_L^{-2} + K_T^{-2} + K_T^{-2})]^{1/2}$. Dimensionless coefficients c_{Fnij} account for the nonlinear increase in form drag with velocity. In this study, L denotes the longitudinal direction parallel to the characteristic direction $\tilde{n}_i$, while T represents any transverse direction perpendicular to $\tilde{n}_i$. It is crucial to clarify that L and T are used to indicate the orientations of the anisotropic porous matrix, as illustrated on the right-hand side of figure 1. Furthermore, coefficient values corresponding to all transverse directions are identical, as required by the transversely isotropic geometry of the porous matrix.

In regions characterised by non-negligible macroscopic velocity gradient, the drag deviation $\phi(R_i - \hat{R}_i)$, the turbulent momentum dispersion $\phi\langle^f u_i^f u_j^f\rangle^f$ and molecular diffusion $2\nu S_{Dij}$ ($S_{Dij} = (\partial_i u_{Dj} + \partial_j u_{Di})/2$) are significantly influenced by the gradient of the macroscopic velocity. To account for anisotropic effects, a second-order pseudo-stress tensor $\tilde{\tau}_{ij}$ should be introduced instead as follows:

$$\phi(R_i - \hat{R}_i) + \nu \partial_{jj}(\phi\langle u_i \rangle^f) - \partial_j(\phi\langle^f u_i^f u_j^f\rangle^f) = \partial_j \tilde{\tau}_{ij}, \quad (2.5)$$

where $\tilde{\tau}_{ij}$ must be symmetric due to angular momentum conservation, and can be made traceless due to incompressibility. The total drag R_i is modelled solely based on the assumption of homogeneity. This assumption may introduce errors, which can be conveniently compensated by adjusting the pseudo-stress tensor. Consequently, $R_i - \hat{R}_i$ is included in the definition of $\tilde{\tau}_{ij}$ in (2.5). Linear mappings from S_{Dij} to $\tilde{\tau}_{ij}$ must be written as

$$\tilde{\tau}_{ij} = \nu_{ijkl} S_{Dkl}, \quad (2.6)$$

with the effective viscosity ν_{ijkl} being a fourth-order tensor with 81 components. The degrees of freedom of ν_{ijkl} can be reduced to 25 using the symmetric and traceless properties of S_{Dij} and $\tilde{\tau}_{ij}$, and further reduced to 3 using the transverse isotropy of porous media. In addition, for the stability of simulation and calibration, the allowed $\tilde{n}_i \tilde{n}_j \tilde{n}_k \tilde{n}_l$ term with an excessive degree of anisotropy is temporarily neglected, which will be justified by modelling accuracy. With further arrangements, the formal expression of $\tilde{\tau}_{ij}$ is

$$\tilde{\tau}_{ij} = 2\nu_L S_{Dij}^{\parallel} + 2\nu_T S_{Dij}^{\perp}, \quad (2.7a)$$

$$S_{Dij}^{\parallel} = (\tilde{n}_i S_{Djk} \tilde{n}_k + \tilde{n}_j S_{Dik} \tilde{n}_k)/2 - (\tilde{n}_i S_{Dkl} \tilde{n}_l) \delta_{ij}/3, \quad S_{Dij}^{\perp} = S_{Dij} - S_{Dij}^{\parallel}. \quad (2.7b)$$

Here, ν_L and ν_T are respectively the effective longitudinal and transversal viscosities. According to the pore-scale prevalence hypothesis (Jin *et al.* 2015), the characteristic length scale of flow in porous media is defined by the pore size s . In this study, we utilise $\sqrt{|K_{ij}|}$ to characterise the flow, as it exhibits a linear relationship with s for a fixed shape of the porous matrix. A Taylor expansion can be expressed with respect to $Re_d = |S_{Dik} K_{kj}|/\nu$, as

$$\nu_L/\nu = \sum_{n=0}^{\infty} c_{LBn} Re_d^n, \quad \nu_T/\nu = \sum_{n=0}^{\infty} c_{TBn} Re_d^n. \quad (2.8)$$

The interphase energy resistance H should be modelled using the effective interfacial convective heat transfer coefficient h , which primarily accounts for the heat transfer $\int_A \alpha \hat{n}_i \partial_i T_f dS/(\phi \Delta V)$ between the fluid and solid phases, expressed as

$$H = -(\phi \Delta V)^{-1} \int_A \alpha \hat{n}_i \partial_i T_f dS \approx \hat{H} = ha(\langle T_f \rangle^f - \langle T_s \rangle^s), \quad (2.9)$$

where α denotes molecular thermal diffusivity, and $a = A/\Delta V$ represents the ratio of the porous matrix surface area to the volume within a REV. From DNS results, two distinct $h \sim Re_K^{\beta}$ scalings are newly identified: $\beta = 2$ in the Darcy regime ($Re_K \lesssim 1$), and $\beta = 0.5$ in the Forchheimer regime ($Re_K \gtrsim 1$). Since the larger/smaller scaling occurs in the low/high Re_K regime correspondingly, simple addition like the Taylor expansion in (2.3) would incorrectly exchange scalings in these regimes, thus a bridge function should be introduced instead:

$$h \sqrt{|K_{ij}|}/\alpha = c_{h0} + c_{h1} Re_K^{1/2} (1 + (Re_K/c_{h2})^{-3/2})^{-1}. \quad (2.10)$$

The angle between fluid velocity and $\tilde{n}$ will affect heat transfer and must be considered:

$$c_{h0} = h_0 \sqrt{|K_{ij}|}/\alpha, \quad c_{h1} = c_{h1ij} u_{Di} u_{Dj} / |u_D|^2, \quad c_{h2} = c_{h2ij} u_{Di} u_{Dj} / |u_D|^2, \quad (2.11a)$$

$$c_{h1ij} = c_{h1L} \tilde{n}_i \tilde{n}_j + c_{h1T} (\delta_{ij} - \tilde{n}_i \tilde{n}_j), \quad c_{h2ij} = c_{h2L} \tilde{n}_i \tilde{n}_j + c_{h2T} (\delta_{ij} - \tilde{n}_i \tilde{n}_j), \quad (2.11b)$$

where h_0 denotes the heat transfer coefficient in the absence of thermal convection. In this study, the Prandtl number $Pr = \nu/\alpha$ is set to 1 for simplicity and is therefore omitted.

Similar to the modelling of momentum dispersion, interphase energy resistance $\phi(H - \hat{H})$, molecular thermal diffusion $\alpha \partial \phi \langle T_f \rangle^f / \partial x_i$ and thermal dispersion $\phi \langle^f u_i^f T_f \rangle^f$ are accounted for by the divergence of a pseudo-flux vector $\tilde{J}_i$ of heat, rather than the commonly employed Laplacian of temperature, to provide a more precise depiction of heterogeneity:

$$\phi(H - \hat{H}) + \alpha \partial_{ii} (\phi \langle T_f \rangle^f) - \partial_i (\phi \langle^f u_i^f T_f \rangle^f) = \partial_i \tilde{J}_i. \quad (2.12)$$

To account for anisotropy, we introduce a second-order anisotropic diffusivity tensor $\tilde{\alpha}_{ij}$ to map the temperature gradient to $\tilde{J}_i$:

$$\tilde{J}_i = \tilde{\alpha}_{ij} \partial_j (\phi \langle T_f \rangle^f), \quad \tilde{\alpha}_{ij} = \alpha_L \tilde{n}_i \tilde{n}_j + \alpha_T (\delta_{ij} - \tilde{n}_i \tilde{n}_j), \quad (2.13a)$$

$$\alpha_L / \alpha = \sum_{n=0}^{\infty} c_{Ln} Re_d^n, \quad \alpha_T / \alpha = \sum_{n=0}^{\infty} c_{Tn} Re_d^n, \quad (2.13b)$$

where α_L and α_T represent the longitudinal and transversal thermal diffusivities.

As shown above, in the closure of macroscopic equations for anisotropic porous media, the tensor approach is used intensively to account for heterogeneity and anisotropy, and the sufficiency and necessity will be shown in the following subsections. The corresponding Taylor expansions are determined based on the specific geometry of porous matrix, as elaborated in § 2.2. For Reynolds number ranges not too large, two leading terms (i.e. the constant and linear terms) of each Taylor expansion can be sufficient.

2.2. Calibration strategy

Without loss of key mechanisms, in all calibration and validation cases we fix the temperature T_s of the porous matrix at zero for simplicity, which can still sufficiently capture the laws of non-equilibrium heat transfer between fluid and solid. This allows the solid phase to function as either a heat sink or a heat source when the surrounding fluid temperature deviates from zero. This approach is analogous to the commonly employed no-slip condition where the solid exhibits velocity zero, thereby acting as a momentum sink or source.

The model coefficients $\hat{R}_i$, $\tilde{\tau}_{ij}$, $\hat{H}$ and $\tilde{J}_i$ are geometric parameters that are independent of the flow conditions and must be determined prior to solving the macroscopic equations. Only the two leading terms of the corresponding Taylor expansions are adopted, which were previously sufficient for modelling purposes (Rao *et al.* 2020; Rao & Zhang 2025b). To determine these coefficients, a schematic geometry of the porous matrix with ratio $d_1/d_2 = 2.0$ and $\phi = 0.81$ is utilised, as illustrated in figure 1. The characteristic direction related to the detailed geometry of the porous matrix is $\tilde{n} = (1 \ 0 \ 0)$. This porous matrix consists of 256 REV's arranged in an $8s \times 8s \times 4s$ configuration, oriented both parallel and perpendicular to the flow direction. To derive a complete set of model coefficients characterising anisotropy, different types of channel flows are necessary, with a unit vector n_w denoting the wall-normal direction, and a driving force g perpendicular to n_w . Periodic boundary conditions are applied in the other two directions parallel to the channel walls.

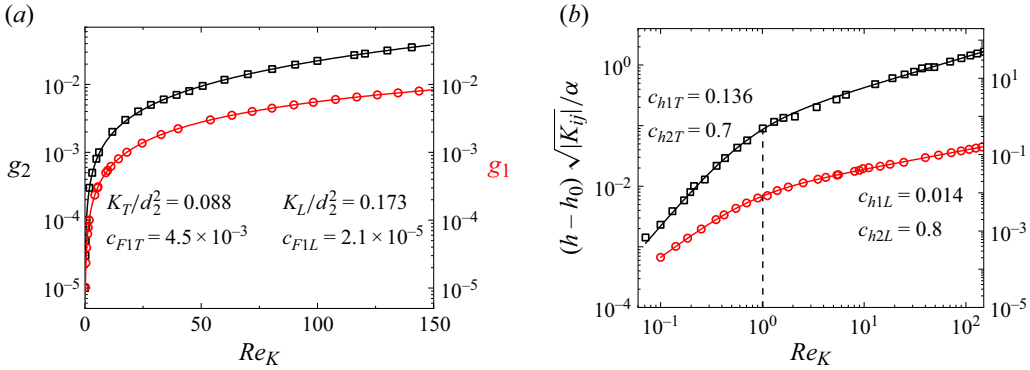


Figure 2. The determination of coefficients by fitting DNS results, $g_2 - n_w = (1 \ 0 \ 0)$ (black, squares, left-hand axis) and $g_1 - n_w = (0 \ 1 \ 0)$ (red, circles, right-hand axis), for (a) $\hat{R}_i$, (b) $\hat{H}$, $c_{h0} = 1.6$.

The heat source is fixed at $\dot{q} = 1$, and the temperature of the channel walls and the porous matrix is fixed at $T_s = 0$. The Reynolds number is varied by adjusting g_i . The configuration is adequate to facilitate the development of turbulent flow and ensure the accurate calculation of the model coefficients.

The finite volume method (FVM) is employed to address the Navier–Stokes equations, while the FVM and the spectral method (SM) are specifically utilised for solving the macroscopic equations. The FVM and SM solvers have already been extensively validated and verified in prior studies (Rao *et al.* 2020; Rao 2022; Rao & Jin 2022; Rao & Zhang 2025a,b), demonstrating the reliability and accuracy of the solvers.

In the core region of the fully developed channel flow, the time-averaged velocity and temperature are strongly homogeneous, and (2.1b,c) can be simplified to

$$\hat{R}_i = g_i, \quad \hat{H} = \dot{q}. \quad (2.14)$$

To determine the coefficients $\hat{R}_i$ and $\hat{H}$, we first specify g_i and subsequently calculate the macroscopic $\phi\langle u_i \rangle^f$ and $\phi\langle T_f \rangle^f$ using DNS results. We then utilise (2.14) to obtain the values of $\hat{R}_i$ and $\hat{H}$. A series of DNS cases is conducted, allowing the model coefficients in (2.3) and (2.10) to be obtained directly by fitting DNS results, as illustrated in figure 2. It can be implied from figure 2 that the total drag $\hat{R}_i$ and the interphase energy resistance $\hat{H}$ exhibit anisotropic properties, with the overall drag and energy resistance coefficients in the transverse direction being larger than those in the longitudinal direction. Regardless of the direction of g_i , h approaches h_0 as $Re_K \rightarrow 0$, as depicted in figure 2(b).

In the first REV close to the wall, which is the only region where momentum and thermal dispersions are non-negligible, the coupled macroscopic equations can be simplified as

$$\partial_j \tilde{\tau}_{ij} = \phi(\hat{R}_i - g_i), \quad \partial_i \tilde{J}_i = \phi(\hat{H} - \dot{q}). \quad (2.15)$$

The errors associated with $\hat{R}_i$ and $\hat{H}$ in the central region may affect the accuracy of the calculated $\tilde{\tau}_{ij}$ and $\tilde{J}_i$. To reduce this uncertainty, the coefficient values of $\hat{R}_i$ and $\hat{H}$ are fine-tuned to ensure that the computed velocity $\phi\langle u_i \rangle^f$ and temperature $\phi\langle T_f \rangle^f$ profiles at the centreline align with the DNS results. In line with our previous research (Rao *et al.* 2020; Rao & Zhang 2025b), the corresponding effective viscosity and thermal diffusivity are determined by fitting DNS results for both low and high Re_K , as shown in figure 3(a,d). And the transition between the two scaling regimes occurs at approximately $Re_K = 1$. The negative effective viscosity shown in figure 3(a) may infer a positive contribution

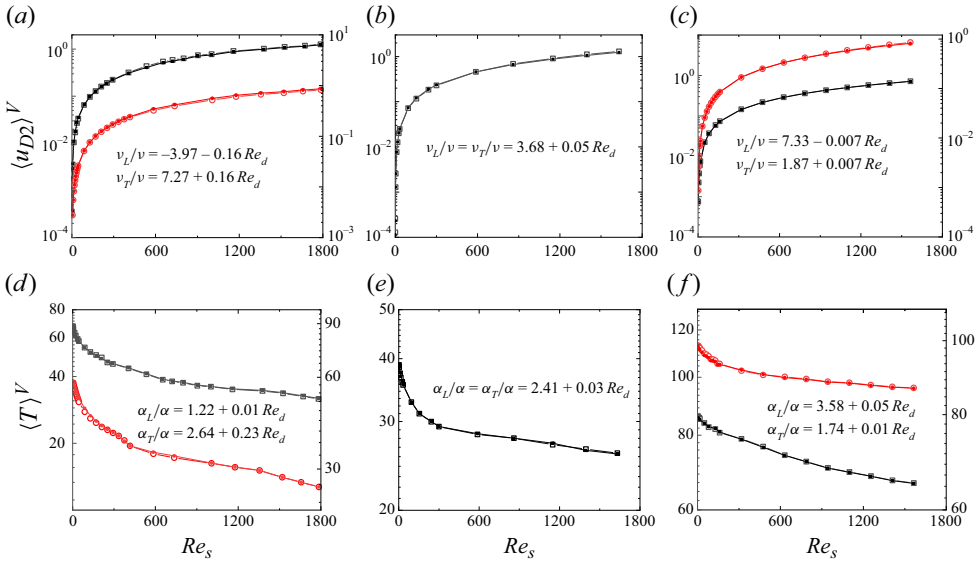


Figure 3. The volume-averaged velocity and temperature in the first REV adjacent to the wall between DNS results (hollow) and model results (solid), for $\phi = 0.81$ and $Pr = 1.0$. Here, $\langle u_{D2} \rangle^V$ is from $g_2 - n_w = (1 \ 0 \ 0)$ cases (black, squares), and $\langle u_{D3} \rangle^V$ is from $g_3 - n_w = (0 \ 1 \ 0)$ cases (red, circles). Plots of (a–c) velocity, and (d–f) temperature, with (a,d) ratio 2.0, (b,e) ratio 1.0, (c,f) ratio 0.5.

to macroscopic kinetic energy, and is already explained in flows without porous media (Bayly & Yakhot 1986; Bhaduri 2021). The negative ν_L in the present work may result from an overestimation of total drag, serving to compensate for this error.

3. Validation and discussion

3.1. Flow in a channel occupied by porous media with one length scale

The macroscopic momentum equation is integrated with the macroscopic energy equation to simulate turbulent forced thermal convection within a channel filled with either anisotropic or isotropic porous media. The schematic arrangement of a porous matrix in figure 1 is utilised. Simulations are conducted for both laminar and turbulent flow regimes. Here, ϕ is fixed at 0.81, with the ratio fixed at 2 or 0.5 representing an anisotropic porous matrix, or ratio being 1 representing the isotropic case. The coefficients $\hat{R}_i$, $\tilde{\tau}_{ij}$, $\hat{H}$ and $\tilde{J}_i$ are determined by fitting DNS results using the methods outlined in § 2.2. Subsequently, the corresponding expressions for $\hat{R}_i$, $\tilde{\tau}_{ij}$, $\hat{H}$ and $\tilde{J}_i$ are directly applied in (2.1) to validate the macroscopic models against additional DNS cases. The pseudo-stress and pseudo-flux significantly influence the velocity and temperature distribution only within the first REV adjacent to the wall boundary. Consequently, the model results are compared with DNS results for the volume-averaged velocity and temperature in the first REV next to the wall, as shown in figure 3.

The model results remain consistent with those from DNS, verifying the sufficiency of the present modelling framework and the truncated Taylor expansions for channel flow in the present range of Re_s . As Re_s increases, the velocity increases, leading to the decrease of temperature. More importantly, when the ratio is larger, equal to and smaller than 1,

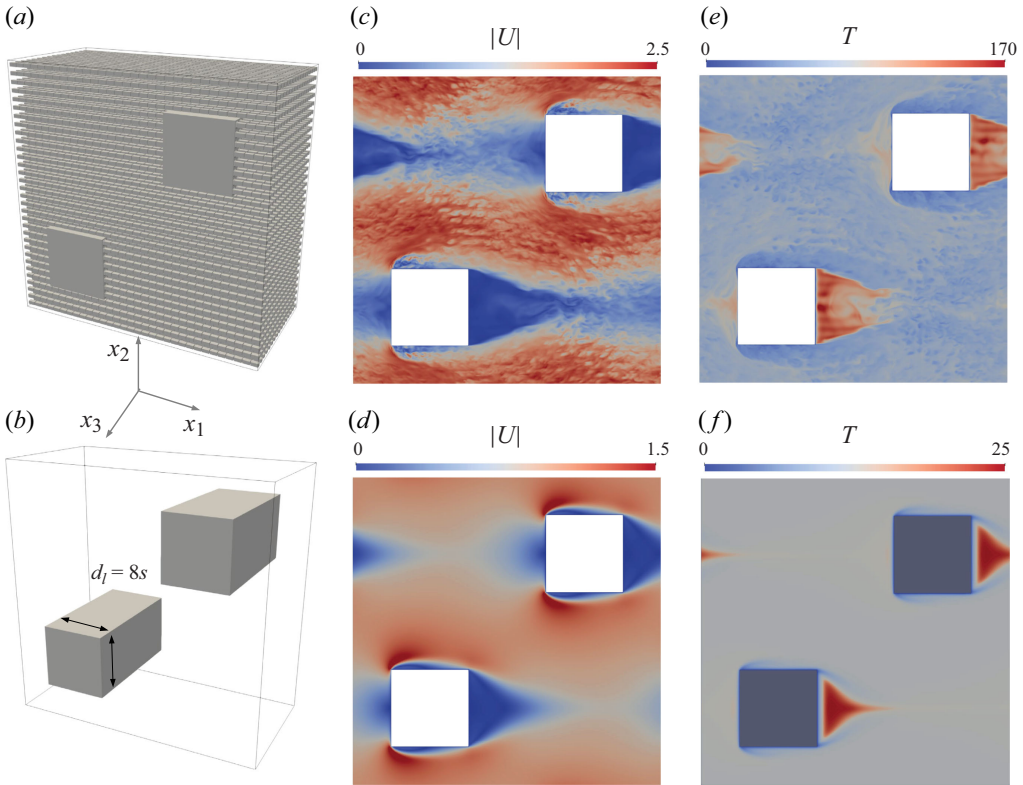


Figure 4. (a,c,e) The DNS, and (b,d,f) the macroscopic simulation, for $Re_s = 793$ and $Pr = 1.0$: (a,b) the schematic of the porous matrix with two length scales; (c,d) the instantaneous velocity magnitude; (e,f) the instantaneous temperature.

the velocity driven by the longitudinal force is larger, equal to and smaller than that driven by the transverse force of the same amplitude, leading to the opposite trend of temperature.

Since the anisotropic properties of momentum and thermal dispersions are intricately linked to the specific characteristics of the porous matrix, it is essential to directly compare the fitted dispersion coefficients across various ratio values. In isotropic porous media with ratio 1 (figure 3b), the transverse components of both momentum and thermal dispersion coefficients (v_T and α_T) are equal to the corresponding longitudinal components (v_L and α_L), and both values are positive, indicating balanced dispersion capability in all directions. In anisotropic porous media with ratio 2 (figure 3a), transverse dispersion coefficients exceed longitudinal values and the values in isotropic porous media of the same porosity. Such anisotropy reflects the enhanced capacity to transport momentum and heat in the transverse direction according to (2.7), (2.8) and (2.13). In contrast, in anisotropic porous media with ratio 0.5 (figure 3c), transverse dispersion coefficients are less than the longitudinal values. Most importantly, the differences indicate the necessity of anisotropic parameters governing momentum and thermal dispersions.

3.2. Flow in a channel occupied by porous media with two length scales

This subsection aims to validate the coupled macroscopic equations in porous media characterised by two distinct length scales. In the DNS, the porous matrices consist of

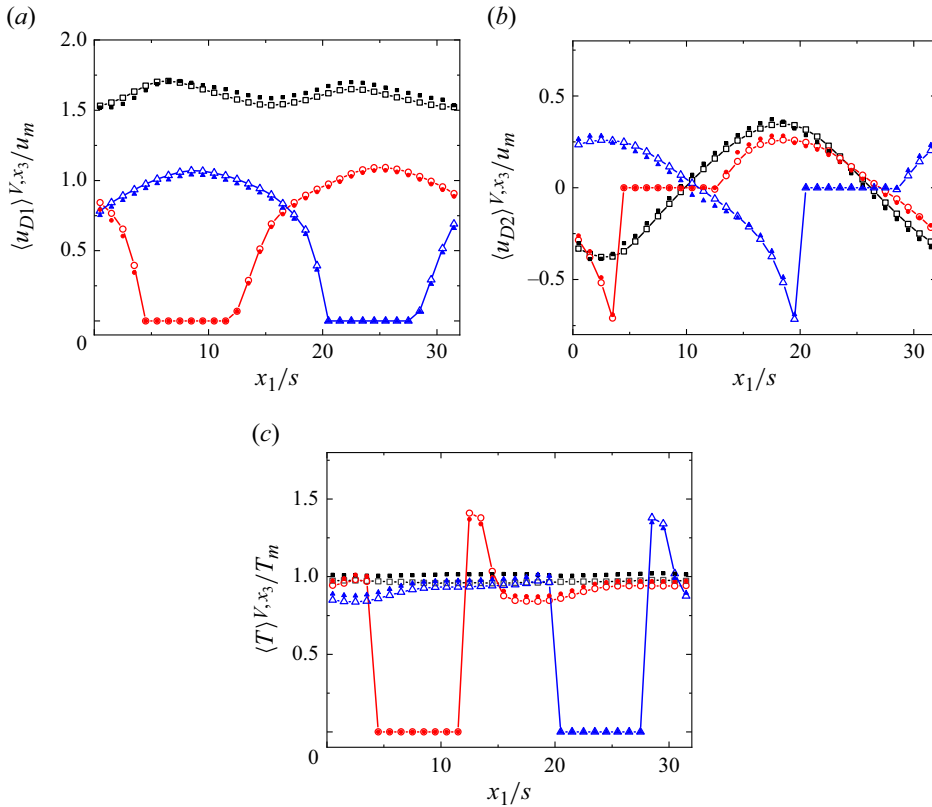


Figure 5. Distributions of $\langle u_{D1} \rangle^{V,x3}$, $\langle u_{D2} \rangle^{V,x3}$ and $\langle T \rangle^{V,x3}$ along the streamwise (x_1) lines, for $Re_s = 793$ and $Pr = 1.0$. Hollow symbols with lines represent DNS results, while solid symbols indicate model results. The three lines are represented by circles (red), squares (black), and triangles (blue) in the sequence, $x_2/s = 4.5$, 16.5 and 27.5: (a) $\langle u_{D1} \rangle^{V,x3}$, (b) $\langle u_{D2} \rangle^{V,x3}$, (c) $\langle T \rangle^{V,x3}$.

cuboids and bars featuring a large element size $d_l = 8s$ and infinite length, as illustrated in figure 4(a). In contrast, the macroscopic simulation utilises porous matrices composed solely of square bars as shown in figure 4(b). The entire simulation domain is configured as $32s \times 32s \times 16s$ for the cuboids with ratio 2, ensuring that turbulent flow can develop fully without being constrained by the computational domain. Periodic boundary conditions are applied in all directions, with the body force g_i imposed in the streamwise (x_1) direction.

The instantaneous velocity and temperature results for both DNS and the macroscopic model are presented in figure 4(c–f). The DNS results show no evidence of large-scale turbulence. Similarly, the model results do not exhibit any turbulence as the turbulence confined to the pore scale has already been filtered through the macroscopic models, which aligns with prior research (Rao & Jin 2022), emphasising that macroscopic models based on the pore-scale prevalence hypothesis effectively account for pore-scale turbulence effects without displaying large-scale turbulence. Figure 5 illustrates the time, REV and spanwise (x_3) averaged components $\langle u_{D1} \rangle^{V,x3}$, $\langle u_{D2} \rangle^{V,x3}$ and $\langle T \rangle^{V,x3}$ along the streamwise (x_1) direction. The three selected streamwise lines are positioned farthest from the bars and intersect the centres of the bars, providing a robust basis for comparing DNS and model results. These quantitative findings indicate a satisfactory accuracy of our macroscopic models. The minor differences observed in REV-averaged velocity between

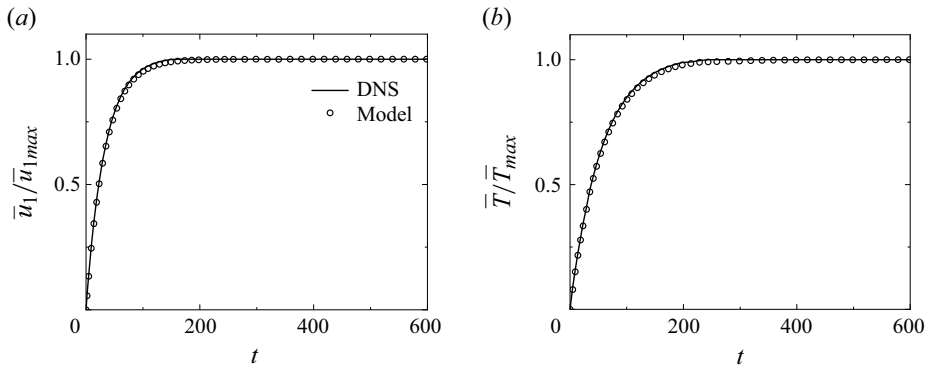


Figure 6. The changes of coupled temperature and velocity versus time, for $Re_s = 793$ and $Pr = 1.0$: (a) velocity, (b) temperature.

the DNS and model results could potentially be further minimised by incorporating additional terms from the Darcy–Forchheimer equation (2.3).

3.3. Further application: prediction of overall time evolution

This subsection highlights the application of coupled macroscopic equations for predicting real-time changes in velocity and temperature. This functionality is critical for effective thermal management, as it necessitates continuous prediction to prevent excessive and uncontrolled temperature increases, thereby ensuring the safety and longevity of devices. The schematic geometry of porous media with two length scales in figure 4(a,b) is utilised. The geometry is considered a complete system, and no macroscopic turbulence emerges, thus the macroscopic equations can be simplified into ordinary differential equations by eliminating the volume V , as

$$\partial_t \bar{u}_i = \phi g_i - \phi \hat{R}_{is}, \quad \partial_t \bar{T} = \phi \dot{q} - \phi \hat{H}_s. \quad (3.1)$$

Here, $\bar{\varphi}$ represents the overall volume average. The effective overall drag $\hat{R}_{is}$ and interphase energy resistance $\hat{H}_s$ can be modelled using the same framework introduced in our previous works as shown in (2.3) and (2.10). The key distinction in this implementation is that we now have fully closed macroscopic equations, which permit direct computation of $\hat{R}_{is}$ and $\hat{H}_s$, leading to a significant reduction in computational costs. This subsection primarily utilises the macroscopic equations (2.1) to track real-time temperature variations. The evolution of volume-averaged velocity and temperature is computed using the Dormand–Prince algorithm implemented as ode45 in MATLAB.

Figure 6 compares the DNS and model results, demonstrating that the coupling model effectively captures the growth of overall velocity and temperature with high consistency. Although a minor discrepancy exists, it may be attributed to the errors in $\hat{R}_{is}$ and $\hat{H}_s$. Notably, the model coefficients are derived only from data obtained under steady or statistically steady flow states. The strong agreement between the model and DNS results suggests that the relaxation time scale for microscopic turbulence to reach an approximate steady state is shorter than the transient growth time scale of mean velocity and temperature. Additionally, it is important to note that temperature responds more slowly than velocity, as it requires time to adjust to variations in velocity. Overall, these coupled macroscopic equations provide a reliable framework for predicting real-time changes in both velocity and temperature.

4. Conclusion

The coupled macroscopic equations governing flow and heat transfer in anisotropic porous media are derived. Anisotropic tensor representations are incorporated to model total drag, momentum dispersion, interphase energy resistance and thermal dispersion, which must be appropriately closed prior to solving the coupled macroscopic equations. To accommodate anisotropic flow fields, the effective physical coefficients in the Darcy–Forchheimer equation that models total drag are adjusted. For the heat transfer coefficient between solid and fluid, this study considers its dependence on flow direction, and identifies $h \sim Re_K^2$ for the Darcy regime, and $h \sim Re_K^{1/2}$ for the Forchheimer regime, with transition at $Re_K \sim 1$. Momentum and thermal dispersion effects, together with the drag and interfacial heat transfer variation induced by heterogeneity, are approximated using a pseudo-stress tensor and a pseudo-flux vector, respectively. The effective viscosity and thermal diffusivity tensors can be decomposed into longitudinal and transverse parts, and approximated using first-order polynomials of Re_d . The effective longitudinal and transverse values of viscosity or thermal diffusivity should be different when the porous media is anisotropic. In isotropic porous media the effective viscosity is consistently positive, whereas in anisotropic porous media the effective viscosity may be negative, highlighting its role in compensating the error of total drag model.

The coupled macroscopic equations have been successfully validated using flows in porous media characterised by either one or two length scales. The results confirm that the proposed models demonstrate high accuracy across a broad range of Reynolds numbers and ratio numbers. The equations account for both anisotropic and isotropic properties, thereby enhancing their applicability to various flow and heat transfer scenarios. Furthermore, the coupled equations are also capable of accurately predicting real-time changes in temperature and velocity, making them particularly advantageous for applications in thermal management. The complete empirical formulas of model coefficients can be derived by further conducting a series of DNS covering various porosities and ratio numbers, which is practical for industrial applications and should be considered in future work.

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REFERENCES

- AGHAEI-JOUYBARI, M., SEO, J.-H., PINTO, S., CATTAFESTA, L., MENEVEAU, C. & MITTAL, R. 2024 Extended Darcy–Forchheimer law including inertial flow deflection effects. *J. Fluid Mech.* **980**, A13.
- BAYLY, B.J. & YAKHOT, V. 1986 Positive- and negative-effective-viscosity phenomena in isotropic and anisotropic Beltrami flows. *Phys. Rev. A* **34** (1), 381.
- BEAVERS, G.S., SPARROW, E.M. & RODENZ, D.E. 1973 Influence of bed size on the flow characteristics and porosity of randomly packed beds of spheres. *J. Appl. Mech.* **40** (3), 655.
- BHADURI, S. 2021 The negative viscosity induces more disturbances in a flow, arXiv preprint arXiv: 2101.05045.
- BRAGA, E.J. & DE LEMOS, M.J.S. 2005 Heat transfer in enclosures having a fixed amount of solid material simulated with heterogeneous and homogeneous models. *Intl J. Heat Mass Transfer* **48** (23–24), 4748–4765.
- BREUGEM, W.P., BOERSMA, B.J. & UITTENBOGAARD, R.E. 2006 The influence of wall permeability on turbulent channel flow. *J. Fluid Mech.* **562**, 35–72.
- DAVIT, Y., DEBENEST, G., WOOD, B.D. & QUINTARD, M. 2010 Modeling non-equilibrium mass transport in biologically reactive porous media. *Adv. Water Resour.* **33** (9), 1075–1093.
- GIVLER, R. & ALTABELLI, S. 1994 A determination of the effective viscosity for the Brinkman–Forchheimer flow model. *J. Fluid Mech.* **258**, 355–370.

- IRMAI, S. 1958 On the theoretical derivation of Darcy and Forchheimer formulas. *EOS Trans. AGU* **39** (4), 702–707.
- JIN, Y., UTH, M.F., KUZNETSOV, A.V. & HERWIG, H. 2015 Numerical investigation of the possibility of macroscopic turbulence in porous media: a direct numerical simulation study. *J. Fluid Mech.* **766**, 76–103.
- KAVIANY, M. 2012 *Principles of Heat Transfer in Porous Media*. Springer Science & Business Media.
- KUZNETSOV, I.A. & KUZNETSOV, A.V. 2017 Using resampling residuals for estimating confidence intervals of the effective viscosity and Forchheimer coefficient. *Transp. Porous Med.* **119**, 451–459.
- LASSEUX, D., VALDÉS-PARADA, F.J. & BELLET, F. 2019 Macroscopic model for unsteady flow in porous media. *J. Fluid Mech.* **862**, 283–311.
- DE LEMOS, M.J.S. 2005 The double-decomposition concept for turbulent transport in porous media. In *Transport Phenomena in Porous Media* (ed. D.B. Ingham & I. Pop), vol. **III**, pp. 1–33. Elsevier Ltd.
- DE LEMOS, M.J.S. 2012a *Turbulence in Porous Media: Modeling and Applications*. Elsevier.
- DE LEMOS, M.J.S. 2012b *Turbulent Impinging Jets into Porous Materials*. Springer Science & Business Media.
- MASUOKA, T. & TAKATSU, Y. 1996 Turbulence model for flow through porous media. *Intl J. Heat Mass Transfer* **39** (13), 2803–2809.
- NAKAYAMA, A., KUWAHARA, F., SUGIYAMA, M. & XU, G. 2001 A two-energy equation model for conduction and convection in porous media. *Intl J. Heat Mass Transfer* **44** (22), 4375–4379.
- NIELD, D.A. & BEJAN, A. 2006 *Convection in Porous Media*. Springer.
- POPE, S.B. 2001 Turbulent flows. *Meas. Sci. Technol.* **12** (11), 2020–2021.
- QUINTARD, M., CHERBLANC, F. & WHITAKER, S. 2001 Dispersion in heterogeneous porous media: one-equation non-equilibrium model. *Transp. Porous Med.* **44**, 181–203.
- RAO, F. 2022 Direct numerical simulation and macroscopic modeling of turbulent flows in porous media. *Thesis*, Universität Bremen, Germany.
- RAO, F. & JIN, Y. 2022 Possibility for survival of macroscopic turbulence in porous media with high porosity. *J. Fluid Mech.* **937**, A17.
- RAO, F., KUZNETSOV, A.V. & JIN, Y. 2020 Numerical modeling of momentum dispersion in porous media based on the pore scale prevalence hypothesis. *Transp. Porous Med.* **133** (2), 271–292.
- RAO, F. & ZHANG, S. 2025a Heat–fluid–solid coupling model for turbulent forced convection within porous media. *Phys. Rev. Fluids* **10** (6), 064502.
- RAO, F. & ZHANG, S. 2025b The in-depth analysis and validation of momentum dispersion modeling based on direct numerical simulation. *Phys. Fluids* **37** (1), 015142.
- REBHI, R., MENNI, Y., LORENZINI, G. & AHMAD, H. 2022 Forced-convection heat transfer in solar collectors and heat exchangers: a review. *J. Adv. Res. Appl. Sci. Engng Technol.* **26** (3), 1–15.
- SAITO, M.B. & DE LEMOS, M.J.S. 2005 Interfacial heat transfer coefficient for non-equilibrium convective transport in porous media. *Intl Commun. Heat Mass Transfer* **32** (5), 666–676.
- SAITO, M.B. & DE LEMOS, M.J.S. 2006 A correlation for interfacial heat transfer coefficient for turbulent flow over an array of square rods. *J. Heat Transfer* **128** (5), 444–452.
- SAMANTRAY, P.K., KARTHIKEYAN, P. & REDDY, K. 2006 Estimating effective thermal conductivity of two-phase materials. *Intl J. Heat Mass Transfer* **49** (21–22), 4209–4219.
- TAM, C.K.W. 1969 The drag on a cloud of spherical particles in low Reynolds number flow. *J. Fluid Mech.* **38** (3), 537–546.
- VALDES-PARADA, F.J., ALBERTO OCHOA-TAPIA, J. & ALVAREZ-RAMIREZ, J. 2007 On the effective viscosity for the Darcy–Brinkman equation. *Physica A* **385** (1), 69–79.
- WHITAKER, S. 1986 Flow in porous media I: a theoretical derivation of Darcy’s law. *Transp. Porous Med.* **1**, 3–25.
- WOOD, B.D., HE, X. & APTE, S.V. 2020 Modeling turbulent flows in porous media. *Annu. Rev. Fluid Mech.* **52**, 171–203.